HandsMen Threads: Elevating the Art of Sophistication in Men's Fashion

User story :

HandsMen Threads, a dynamic organization in the men's fashion industry, is implementing a Salesforce-based platform to modernize data management, streamline business operations, and elevate customer relations. As a Sales Representative, I want to track customer purchases, preference profiles, and order status in one central application, so that I can provide personalized styling recommendations and deliver a seamless sales experience. As a Warehouse / Inventory Manager, I want automated alerts whenever product stock levels drop below thresholds, so that I can initiate timely restocking and prevent stockout delays. As a Customer Experience Manager, I want post-order email updates and dynamic loyalty program status updates, so that customers stay engaged and rewarded for repeat business. As a Finance / Operations Administrator, I want scheduled nightly bulk updates to capture large order adjustments and update financial records automatically, so that daily financial and inventory reports are always accurate and ready for management decision-making.

1. INTRODUCTION

1.1 Project Overview :

HandsMen Threads is a Salesforce-based project designed to simplify men's fashion retail operations by centralizing customer details, product information, order records, inventory data, and marketing campaigns. The system uses custom objects, fields, lookup relationships, master-detail relationships, and formula fields to organize business data efficiently. A dedicated Salesforce Lightning application provides a unified interface for accessing and managing these records.

The project incorporates validation rules and Apex triggers to maintain data accuracy, while email templates, email alerts, and record-triggered flows support order confirmation and low-stock notifications. A scheduled flow is designed to update customer loyalty statuses based on purchase totals, and a batch Apex class supports scheduled inventory processing. Profiles, roles, and permission sets help manage user access, improving operational efficiency, data consistency, and customer service.

1.2 Purpose:

- To centralize customer, product, order, inventory, and marketing campaign information in Salesforce.
- To simplify the management of customer details and order records through a dedicated Salesforce Lightning application.
- Automated To maintain data accuracy using validation rules, formula fields, and Apex triggers and To automate order confirmation emails and low-stock alerts using email alerts and Salesforce flows.
- To manage user access securely through profiles, roles, and permission sets.

- To improve operational efficiency, data consistency, and customer service in retail operations.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1 Sales Representative	Manage customer details, track orders, and provide better customer service.	Customer information and order details can be difficult to manage without a centralized system.	Customer and order records need to be maintained and updated systematically.	Frustrated by manual work and concerned about delays in handling customer orders.
PS-2 Inventory Manager	Monitor product stock levels and maintain accurate inventory records.	Low-stock products may not be identified in time.	Inventory tracking requires regular monitoring and timely updates.	Concerned about stock shortages and possible delays in fulfilling orders.
PS-3 Customer Experience Manager	Improve customer retention and manage loyalty statuses effectively.	Updating customer loyalty statuses manually can be time-consuming.	Customer purchase totals must be reviewed to determine the appropriate loyalty status.	Concerned about delays in recognizing loyal customers and maintaining customer satisfaction.
PS-4 Administrator / Operations Lead	Maintain accurate customer, product, order, and inventory data while ensuring smooth business operations.	Manual data entry and inconsistent updates can lead to errors.	Business records require proper validation, access control, and regular synchronization.	Overwhelmed by repetitive tasks and uncertain about data accuracy.

2.2 Empathy Map Canvas

User Persona: Sales Representative & Inventory Manager

- **Think & Feel:** Concerned about incorrect order details and low product stock. Eager to provide better customer service and maintain accurate inventory records.
- **Hear:** Customer inquiries about order confirmation and product availability, along with management requests for accurate sales and inventory information.
- **See:** Customer records, product details, order information, and inventory stock levels maintained within the Salesforce HandsMen Threads application.
- **Say & Do:** Updates customer and order records, checks product availability, monitors inventory levels, and uses automated email notifications to support order confirmation and low-stock alerts.

2.3 Brainstorming & Idea Prioritization

Step 1: Brainstormed Ideas

- **Centralized Data Management**: Use Salesforce custom objects to maintain customer, product, order, inventory, and marketing campaign records in one place.Third-Party Standalone Portal: A custom e-commerce portal integrated loosely via external webhooks.

- **Salesforce Lightning Application**: Develop the HandsMen Threads application with customized navigation items for easy access to business records.
- **Automated Order Confirmation:** Use email templates, email alerts, and record-triggered flows to send order confirmation notifications.
- **Data Validation and Error Prevention**: Implement validation rules and Apex triggers to enforce business rules and maintain accurate order information..
- **User Access Management**: Configure profiles, roles, and permission sets to control access to Salesforce objects and records.

Step 2: Grouping

Group	Ideas	Description
Manual solutions	Manual Customer Records, Spreadsheet-Based Inventory Tracking	Easy to start, but manual data entry can lead to errors, duplicate records, and delays in updating customer and inventory information.
Partially automated solutions	Formula Fields, Validation Rules, Individual Email Templates	Improves data accuracy and supports specific tasks, but requires additional automation to manage complete business workflows..
Fully automated cloud solutions	Salesforce HandsMen Threads App, Record-Triggered Flows, Email Alerts, Scheduled Flows, Batch Apex Jobs	Provides a centralized cloud platform for managing customer, product, order, inventory, and marketing campaign records with automated notifications, data validation, and scheduled processing.

Step 3: Idea Prioritization

Each idea is rated High, Medium, or Low based on its impact, feasibility, and cost efficiency.

Idea	Impact	Feasibility	Cost Efficiency
Manual Customer and Inventory Records.	Low	High	High
Salesforce HandsMen Threads Application	High	High	High
Validation Rules and Apex Triggers	High	High	High
Automated Order Confirmation Emails	High	High	High
Automated Low-Stock Alerts	High	High	High
Scheduled Loyalty Status Updates	High	Medium	High
Profiles, Roles, and Permission Sets	High	High	High

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stage 1: Customer Management & Order Creation

- User Actions: Sales Representative creates customer records, checks customer details, and enters order information into Salesforce.
- Touchpoints: HandsMen Threads Salesforce Lightning App, HandsMen Customer, HandsMen Order.
- Pain Points: Missing customer information or incorrect order details.
- Solution: Custom objects, lookup relationships, formula fields, and validation rules help organize records and maintain data accuracy.
- Emotion: Confident.

Stage 2: Order Confirmation & Notification

- User Actions: Sales Representative updates the order status to "Confirmed".
- Touchpoints: HandsMen Order record, Order Confirmation Flow, Email Alert, Customer Inbox.
- Pain Points: Delayed order confirmation and uncertainty about order status.
- Solution: A record-triggered flow sends an automated order confirmation email when the order status changes to "Confirmed".
- Emotion: Reassured.

Stage 3: Customer Loyalty Status Update

- User Actions: The system evaluates customer purchase totals and assigns an appropriate loyalty status.
- Touchpoints: HandsMen Customer records, Scheduled Flow, Loyalty Status field.
- Pain Points: Manual loyalty calculations may cause errors or delays in updating customer status.
- Solution: A scheduled flow evaluates total purchases and assigns Gold, Silver, or Bronze loyalty status based on the configured purchase thresholds.
- Emotion: Appreciative.

Stage 4: Inventory Monitoring & Low-Stock Alerts

- User Actions: Inventory Manager monitors stock quantities and identifies products with low inventory.
- Touchpoints: Inventory records, Stock Status formula field, Stock Alert Flow, Email Alert.

- Pain Points: Low stock may not be identified early, potentially affecting order fulfillment.
- Solution: A formula field identifies stock status, and a record-triggered flow checks whether stock quantity falls below 5 units and sends an alert to the Inventory Manager.
- Emotion: Confident.

Stage 5: Scheduled Inventory Processing

- User Actions: The system runs a scheduled Batch Apex job to process eligible inventory-related product records.
- Touchpoints: InventoryBatchJob, Scheduled Apex Job, HandsMen Product records.
- Pain Points: Processing multiple records manually can be time-consuming and may delay inventory updates.
- Solution: Batch Apex processes qualifying records in groups, while a scheduled job runs the process automatically at the configured time.
- Emotion: Satisfied.

3.2 Solution Requirements

Functional Requirements

- FR-1: Data Modeling and User Interface: Create custom objects for HandsMen Customer, HandsMen Product, HandsMen Order, Inventory, and Marketing Campaign, with a Lightning application and custom tabs.
- FR-2: Automated Order Confirmation: Send automated confirmation emails when an order status changes to "Confirmed" using email templates, email alerts, and record-triggered flows.
- FR-3: Customer Loyalty Management: Use a scheduled flow to update customer loyalty statuses to Gold, Silver, or Bronze based on total purchase amounts.
- FR-4: Proactive Stock Alerts: Send email notifications to the Inventory Manager when stock quantity falls below 5 units using a record-triggered flow.
- FR-5: Scheduled Inventory Processing: Use Batch Apex and scheduled jobs to process eligible product records and perform inventory-related updates automatically.

Non-Functional Requirements

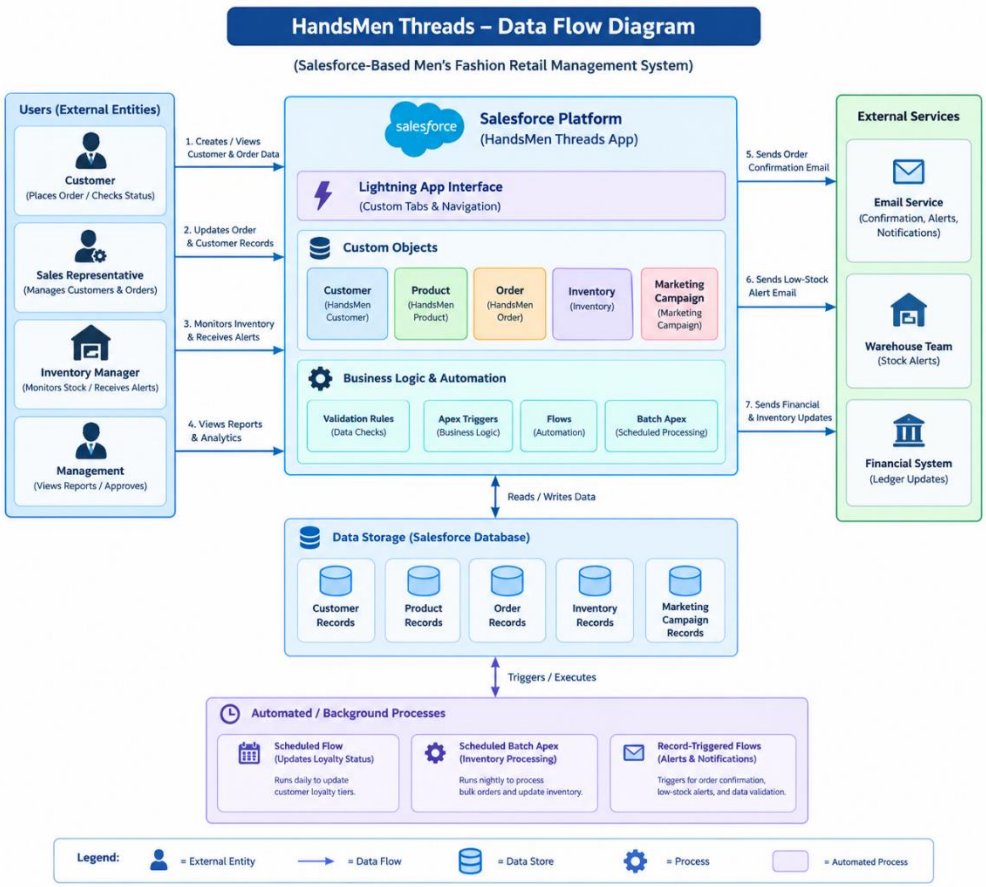
- NFR-1: Security: Use profiles, roles, and permission sets to control access to Salesforce objects and records.
- NFR-2: Usability: Provide a simple and organized Lightning application with clear navigation for easy record management.
- NFR-3: Performance: Use Batch Apex to process records in manageable groups and support efficient bulk processing.

- NFR-4: Scalability: Use Salesforce cloud infrastructure to support increasing volumes of customer, product, order, and inventory records.

- NFR-5: Data Integrity: Apply validation rules, formula fields, and Apex triggers to prevent invalid data and enforce business rules.

3.3 Data Flow Diagram

This is the Level-1 data flow diagram for the main steps of the system.



Here is how the diagram maps to the steps:

- **Step 1: Input and Storage:**

  The Sales Representative enters customer and order details through the Salesforce application. Validation rules check the entered data before it is saved in the relevant customer, order, and product records.

- **Step 2: Automation Execution:**

  When an order's status changes to Confirmed, the record-triggered flow executes the configured automation process.

- **Step 3: Customer Engagement:**

  System retrieves customer contact info and sends an Order Confirmation Email.

- **Step 4: Loyalty Calculation:**

  System aggregates purchase totals and updates the customer's loyalty tier.

- **Step 5: Inventory Monitoring:**

  The system monitors inventory quantities. When the stock quantity falls below 5, the stock alert flow sends an email notification to the Inventory Manager.

- **Step 6: Scheduled Processing:**

  A scheduled Batch Apex job processes eligible product records in batches and updates inventory quantities according to the implemented logic. The job is scheduled to run at midnight.

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem–Solution Mapping

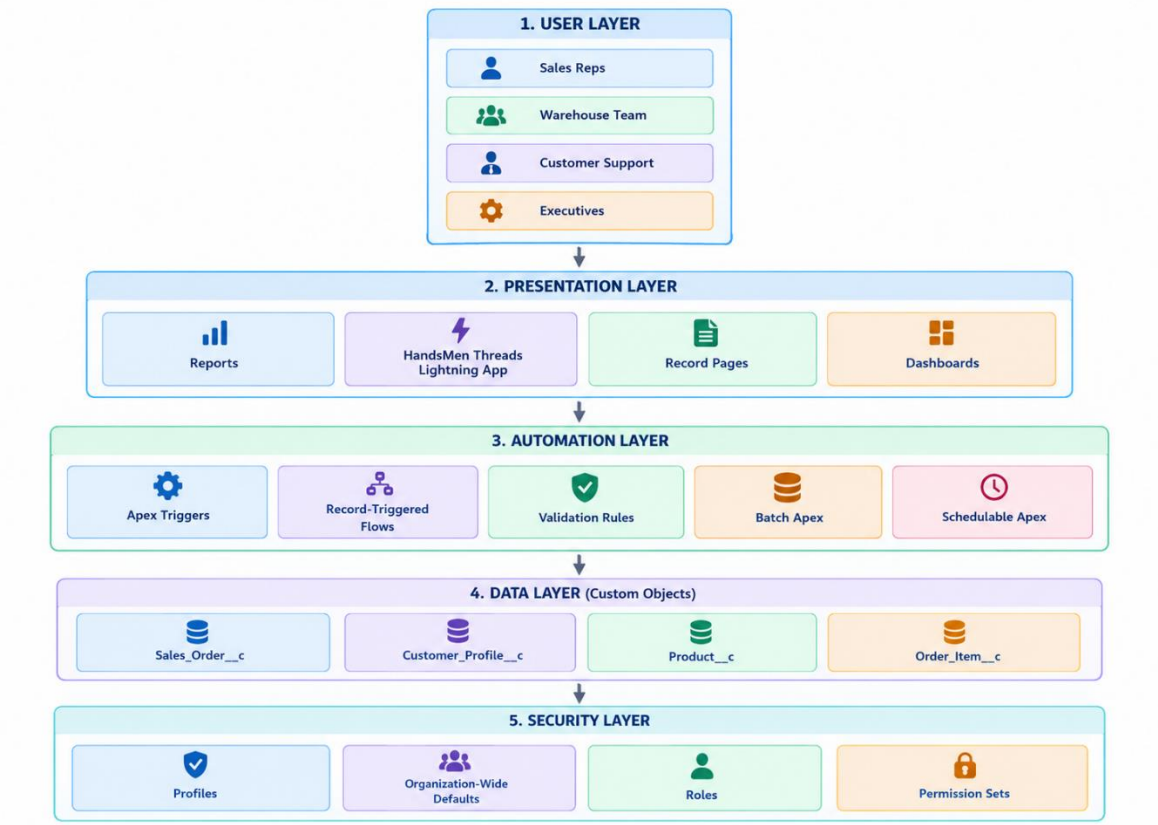
Problem (User Behavior / Constraint)	HandsMen Threads Solution	Salesforce-Native Capability
Scattered customer & order data	Centralized fashion data model	Custom Objects: HandsMen Customer, HandsMen Product, HandsMen Order, Inventory, and Marketing Campaign
Data entry errors & invalid formats	Validation of order details before saving	Validation Rules, Field Constraints, and Apex Triggers
Missing post-order communications	Automated order confirmation emails	Email Templates, Email Alerts, and Record-Triggered Flows
Delayed loyalty status assignment	Scheduled calculation of customer loyalty status based on purchase totals	Scheduled Flows
Unannounced stockouts	Automated warehouse stock alerts	Record-Triggered Flows (Stock Threshold < 5)
Unauthorized access to sensitive records	Controlled access based on user permissions	Profiles, permission sets, roles, and sharing rules
Time-consuming bulk inventory processing	Scheduled processing of eligible product records	Batch Apex and Scheduled Apex Jobs
Inconsistent data presentation	Automatic calculation of full names and stock status	Formula Fields

4.2 Proposed Solution

Parameter	Description
Problem	Disjointed sales tracking, manual inventory checks, delayed customer communication, and potential system slowdown during bulk processing.
Solution	A custom Salesforce application unifying data management, UI validation, automated workflows, and scheduled asynchronous processing.
Novelty	End-to-end integration of fashion order management with proactive inventory threshold alerts and automated nightly batch operations on one platform.
Social Impact	Professionalized operations, reduced material waste from stock inaccuracies, and higher workplace productivity.
Revenue Model	Enterprise apparel management framework with reduced operational overhead and improved client retention.
Scalability	Cloud-native platform capable of onboarding multi-store retail branches without core architectural re-engineering.

4.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers. Users interact through the Lightning interface, automation processes their requests, and the results are stored in custom objects. A security layer controls access at every level.



Explanation of Solution Architecture Layers

1. Users Layer: Provides role-based interaction for Sales Representatives, Inventory Managers, Customer Support, and Management to perform their respective tasks.

2. Presentation Layer: Uses the HandsMen Threads Lightning App, custom tabs, record pages, reports, and dashboards to provide an organized interface for managing and viewing business data.

3. Automation Layer: Uses Record-Triggered Flows for order confirmation and stock alerts, Apex Triggers for order validation, Validation Rules for data accuracy, and Batch Apex with scheduled execution for inventory processing.

4. Data Layer: Stores customer, product, order, inventory, and marketing campaign information using Salesforce custom objects, lookup relationships, and master-detail relationships.

5. Security Layer: Controls user access through profiles, roles, and permission sets to help protect records and ensure users have appropriate permissions.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project follows the **Agile methodology**. Work is organized as **Epics → User Stories → Story Points** and delivered in **sprints**. Velocity from completed sprints is used to estimate the remaining work.

Product Backlog and Sprint Schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	Set up developer edition org and developer environment.	1	High	Member1
Sprint-1	Data Modeling	USN-2	Create custom objects (Customer_Profile, Product, Sales_Order, Order_Item).	3	High	Member2
Sprint-2	Data Modeling	USN-3	Create custom tabs and navigation styles for all objects.	3	High	Member4
Sprint-2	Data Modeling	USN-4	Build custom fields (Picklists, Formulas, Currency, Lookup/Master-Detail).	5	High	Member1
Sprint-2	App Building	USN-5	Configure HandsMen Threads Lightning App and assign profile access.	3	High	Member3
Sprint-2	Data Modeling	USN-6	Define relationships (Master-Detail for Order-Items, Lookups for Orders)	5	High	Member2
Sprint-2	Page Layouts	USN-7	Design tailored page layouts for Sales and Warehouse teams.	5	High	Member1
Sprint-3	Automation	USN-8	Implement Validation Rules for UI data accuracy.	3	High	Member4
Sprint-3	Automation	USN-9	Create email templates and Record-Triggered Flow for order confirmations.	3	High	Member3

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-3	Automation	USN-10	Build Record-Triggered Flow for dynamic loyalty program status updates.	5	High	Member1
Sprint-4	Automation	USN-11	Build Record-Triggered Flow for stock alerts when quantity drops below 5.	3	High	Member2
Sprint-5	Development	USN-12	Develop Apex Triggers for complex data verification.	5	High	Member2
Sprint-5	Development	USN-13	Develop Asynchronous / Batch Apex for scheduled bulk order updates at midnight.	5	High	Member4
Sprint-5	Security	USN-14	Configure roles, profiles, and permission sets for security setup.	5	High	Member 3
Sprint-6	Analytics	USN-15	Create custom reports and operational dashboards.	4	High	Member 1

Epic Summary

Epic	Stories	Story Points
Developer Setup	USN-1	1
Data Modeling & Design	USN-2 to USN-7	24
Automation	USN-8 to USN-11	14
Apex Development	USN-12 to USN-13	10
Security	USN-14	5
Analytics & Reports	USN-15	4
Total	**15 stories**	**58**

Sprint Schedule

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date
Sprint-1	4	7 days	5 Oct 2026	11 Oct 2026	4	11 Oct 2026
Sprint-2	21	7 days	12 Oct 2026	18 Oct 2026	21	18 Oct 2026
Sprint-3	11	7 days	19 Oct 2026	25 Oct 2026	11	25 Oct 2026
Sprint-4	8	7 days	26 Oct 2026	1 Nov 2026	8	1 Nov 2026
Sprint-5	10	7 days	2 Nov 2026	8 Nov 2026	10	8 Nov 2026
Sprint-6	4	7 days	9 Nov 2026	15 Nov 2026	4	15 Nov 2026
Total	**58**	**42 days**			**58**	

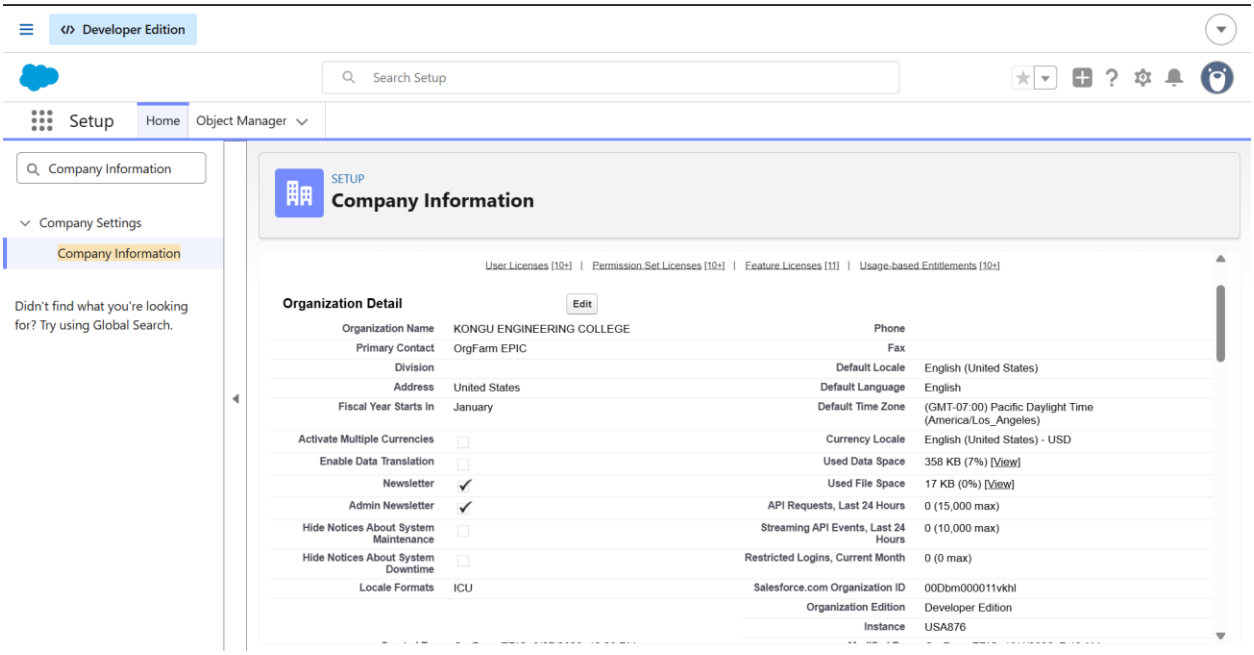
Team Workload

Member	Stories	Total Story Points
Member1	USN-1, USN-4, USN-7, USN-10, USN-15	20
Member2	USN-2, USN-6, USN-11, USN-12	16
Member3	USN-5, USN-9, USN-14	11
Member4	USN-3, USN-8, USN-13	11

6. Project Development Phase

Milestone 1: Salesforce Developer Account Creation

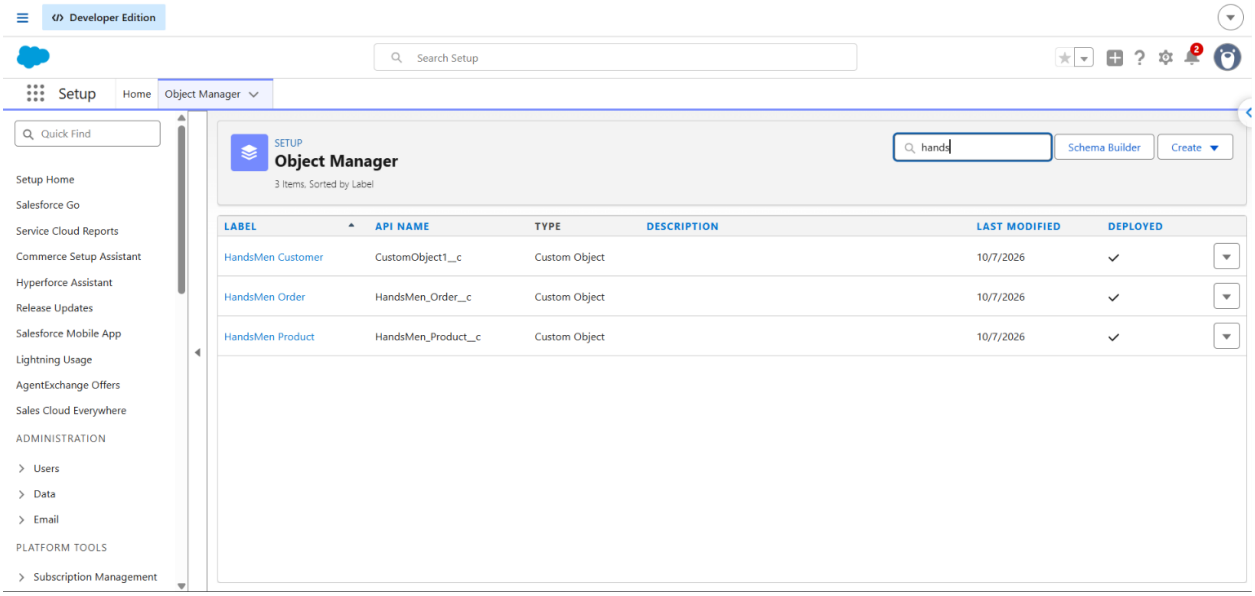
- **Definition:** A free Salesforce Developer Edition org serves as the environment for building and testing the application.
- **Key Steps**: Sign up at developer.salesforce.com, verify email, log in, and configure administrative preferences.



Milestone 2: Custom Object Creation

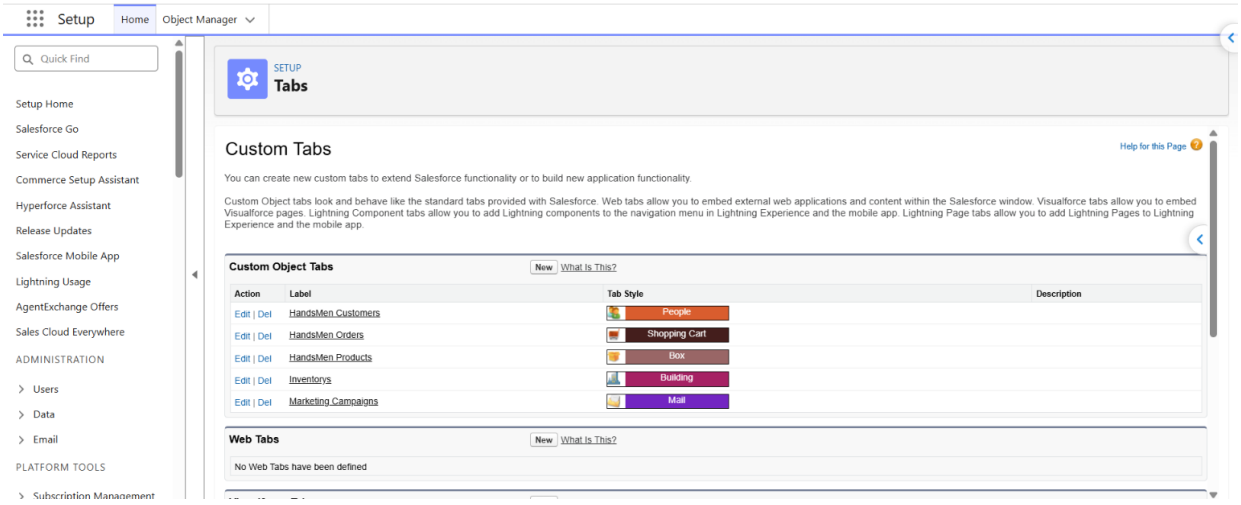
- **Definition**: Establishing the main custom objects required to manage customer details, product information, sales orders, and order items in Salesforce.

- **Key Steps:** Open Object Manager, create each custom object with appropriate labels and singular/plural names, configure record name formats, and enable reporting wherever required.



Milestone 3: Tabs Creation

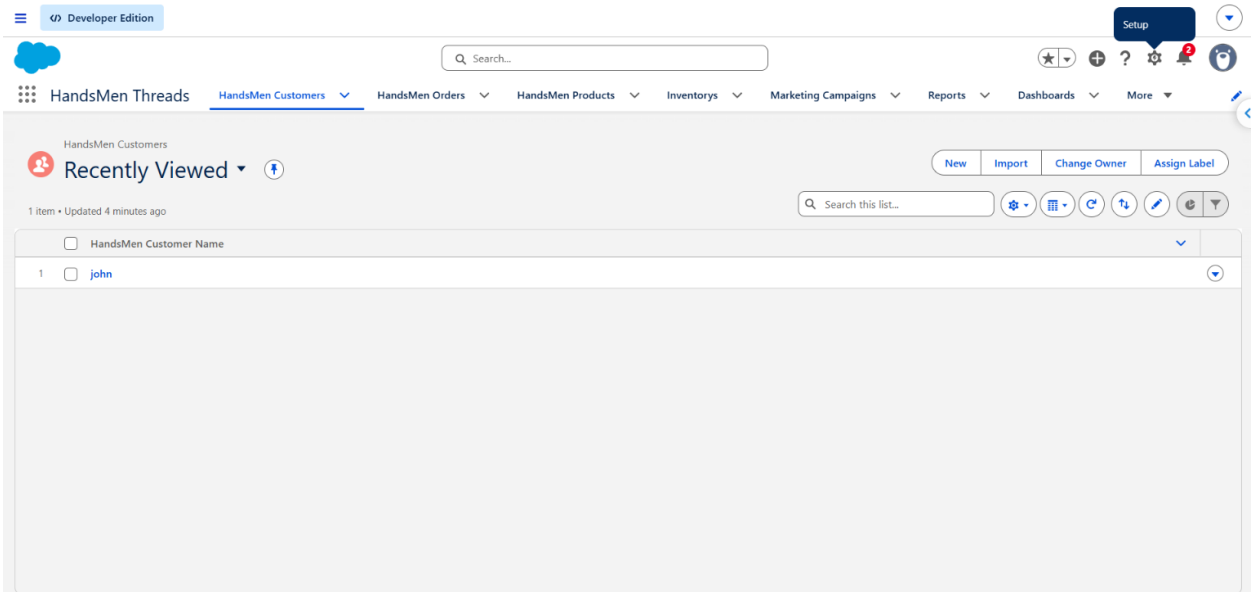
- **Definition:** Creating dedicated navigation tabs to provide easy access to the main data objects within the Salesforce application.
- **Key Steps:** Navigate to Setup, create custom object tabs, select suitable tab styles, and configure tab visibility based on user profiles.



Milestone 4: Lightning App Creation

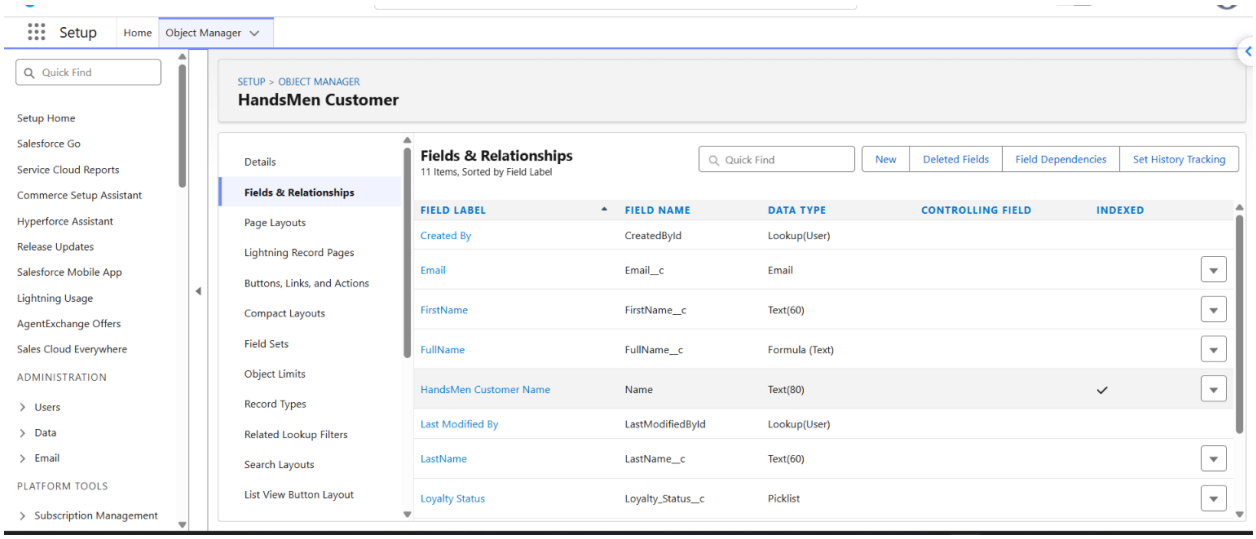
- **Definition:** Developing a customized Salesforce Lightning application named “HandsMen Threads” that integrates essential navigation tabs and business management features in one place.

- **Key Steps:** Open App Manager, create a Lightning application with standard navigation, apply suitable branding, add the required custom tabs and utility items, and configure app access for the appropriate user profiles.



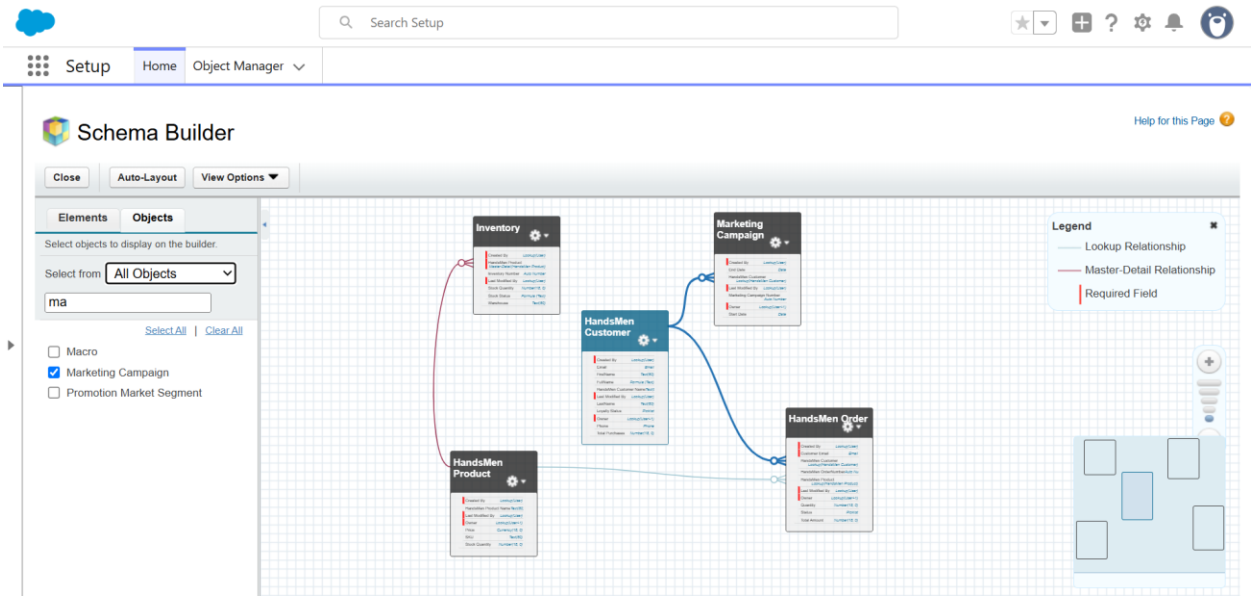
Milestone 5: Fields Creation

- **Definition:** Configuring custom fields to store essential information such as stock quantities, product prices, order statuses, and customer details across Salesforce objects.
- **Key Steps:** Create appropriate custom fields using data types such as Currency, Number, Text, Picklist, and Formula, and configure the required field properties and user interface settings.



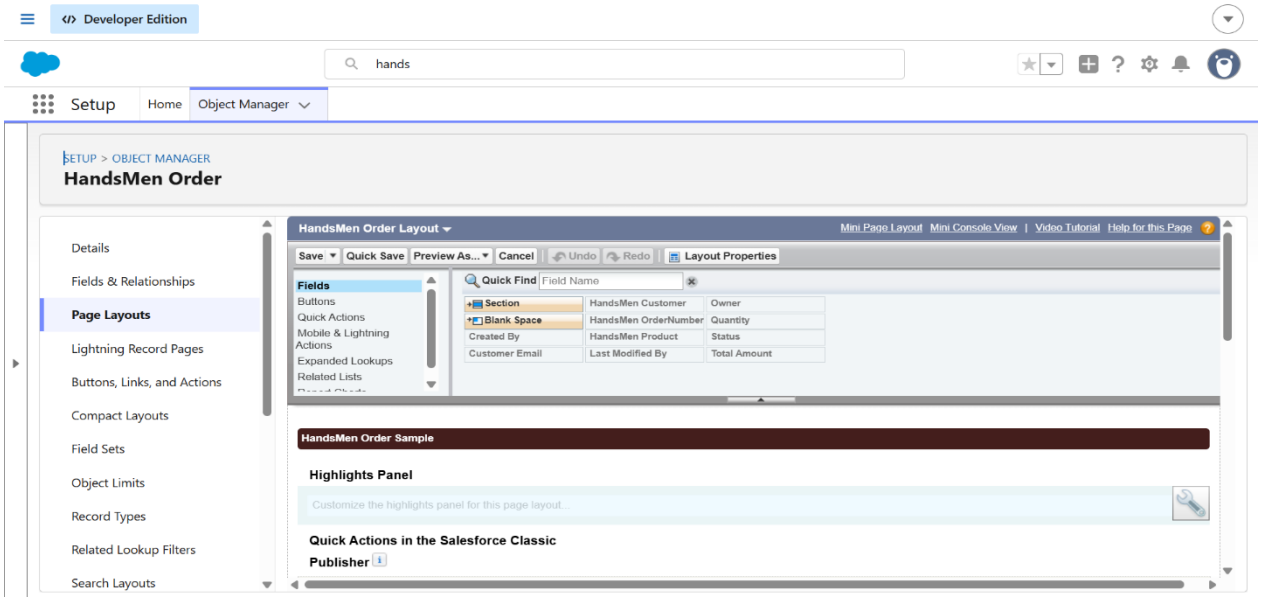
Milestone 6: Create Relationships

- **Definition:** Linking records across objects to structure order lineups and customer histories.
- **Key Steps**: Build Master-Detail relationships between Order_Item__c and Sales_Order__c, and Lookup relationships connecting orders to customers and products.



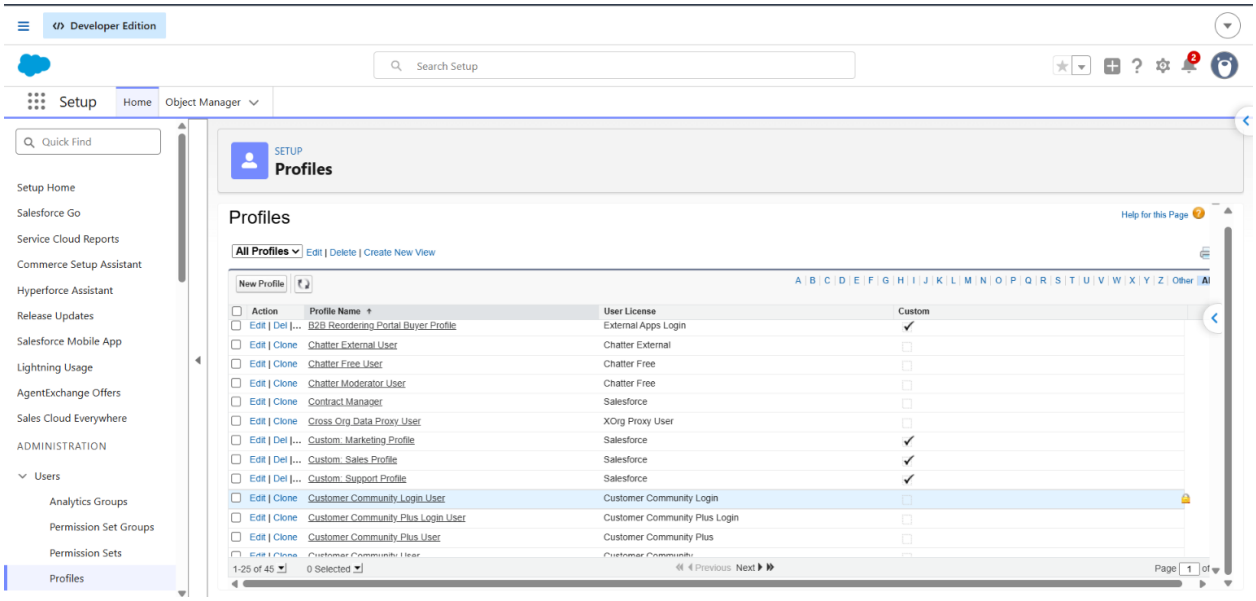
Milestone 7: Page Layout Customization

- **Definition:** Tailoring layout screens to display relevant fields clearly based on user roles.
- **Key Steps:** Group fields into organized sections, adjust field accessibility settings, and place related list views on primary record pages.



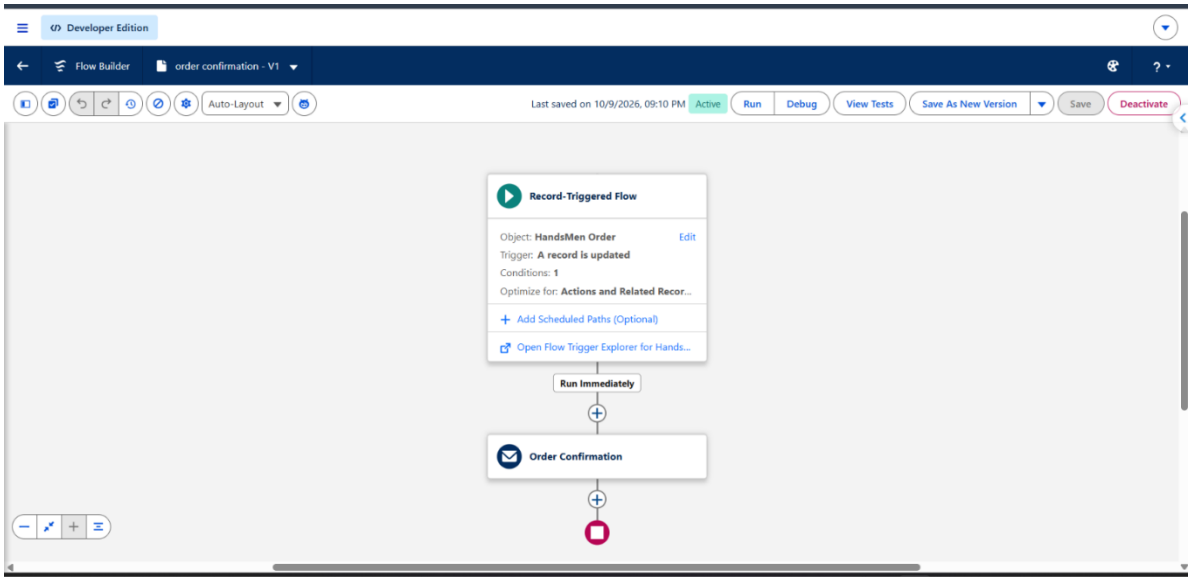
Milestone 8: Profile Creation

- **Definition:** Creating customized Salesforce profiles to manage user access and permissions according to different business responsibilities, including marketing, sales, and customer support.
- **Key Steps:** Navigate to Setup → Users → Profiles and clone a suitable existing profile to create Marketing, Sales, and Support profiles.



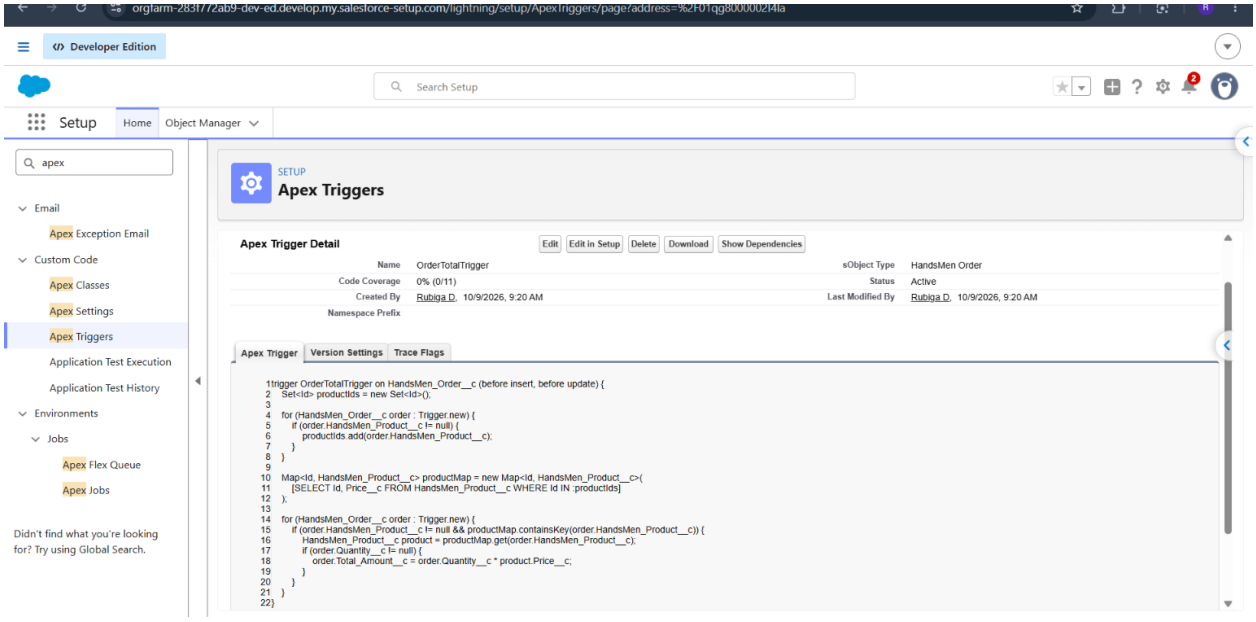
Milestone 9: Record-Triggered Flows & Email Templates

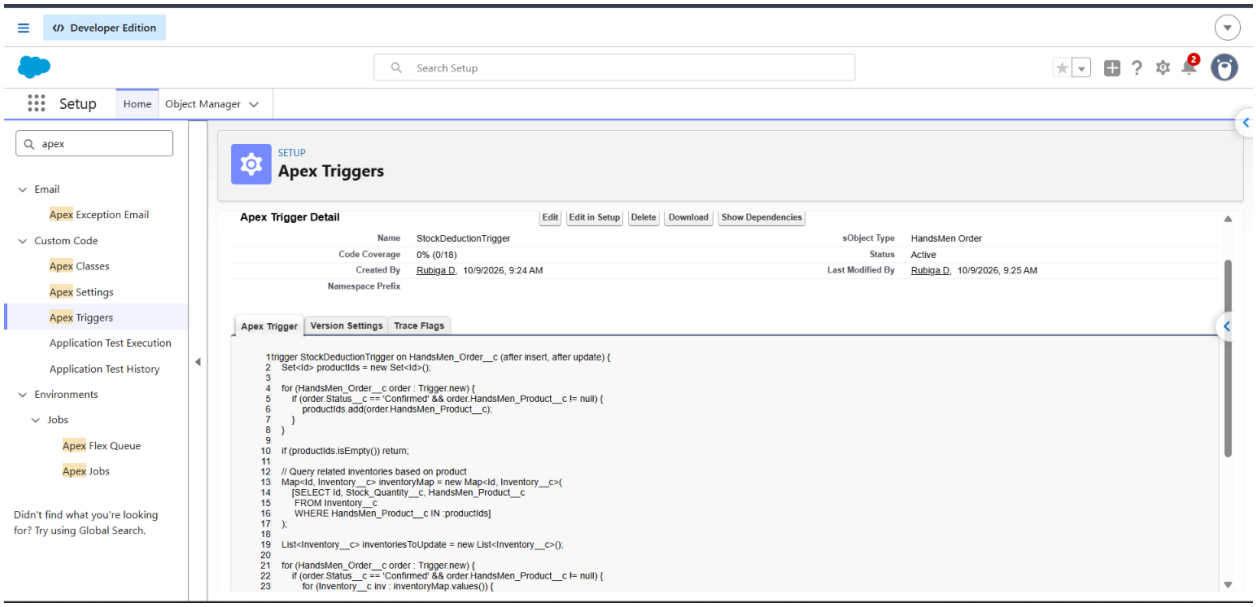
- **Definition**: Automating post-order emails, stock alerts, and loyalty tier updates without custom code.
- **Key Steps**: Build email templates. Configure Record-Triggered Flows on Sales_Order__c and Product__c for automated customer order emails, loyalty status updates, and low-stock alerts (< 5 units) to warehouse staff.



Milestone 10: Apex Triggers logic

- **Definition:** Custom Apex logic for advanced business rules and validation constraints exceeding declarative capabilities.
- **Key Steps:** Implement Apex Triggers on order insertion/update events, ensuring governor-limit compliant bulk handling with test class execution.





Milestone 14: Conclusion

Project Summary

The HandsMen Threads project establishes a unified, cloud-based management platform using Salesforce to replace fragmented manual record-keeping. By integrating data modeling, automated email notifications, customer loyalty management, low-stock alerts, and scheduled batch processing, the system streamlines retail operations and improves overall data management.

Outcomes and Benefits

- **Operational Efficiency:** Automated order confirmation emails and inventory alerts reduce manual effort and simplify routine operations.
- **Improved Data Accuracy:** Validation Rules and Apex Triggers help prevent invalid data entries and maintain data consistency.
- **Proactive Inventory Management:** Automated notifications for stock quantities below five units help identify potential stock shortages.
- **Efficient Bulk Processing:** Scheduled Batch Apex jobs process eligible records in batches, supporting background inventory operations.
- **Scalability and Security:** Custom profiles, roles, and permission sets help organize user access and support future business growth.

Possible Future Enhancements

- **Customer Self-Service Portal**: Develop a Salesforce Experience Cloud portal where customers can track orders, view loyalty status, and update their profiles.
- **Omnichannel Notifications:** Integrate SMS and messaging services, such as WhatsApp Business API, to provide timely order and delivery updates.
- **E-Commerce Integration:** Connect the Salesforce platform with online stores through REST APIs to synchronize product and inventory information.
- **AI-Driven Analytics:** Explore Salesforce Einstein capabilities to forecast apparel demand and provide personalized product recommendations.

Closing Statement

The HandsMen Threads Care and Fashion Management System provides a centralized foundation for managing retail operations through Salesforce. By combining structured data management, automation, security controls, and scheduled processing, the project aims to improve operational efficiency, data reliability, and customer service while providing opportunities for future enhancements.